Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: R Doornenbal Dairy #1

Physical address of dairy:

Number and Street	City	County	Zip Code
19116 Lone Tree RD	Escalon	San Joaquin	95320

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1940

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X205-X060-XX07-XXXX X205-X060-XX08-XXXX X205-X110-XX12-XXXX

B. OPERATORS

Doornenbal, Rien

Operator name: Doornenbal, Rien

Telephone no.: (209) 838-7108 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
19116 Lonetree RD	Escalon	CA	95320

This operator is responsible for paying permit fees.

C. OWNERS

Doornenbal, Rien

Legal owner name: Doornenbal, Rien

Telephone no.: (209) 838-7108 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
19116 Lonetree RD	Escalon	CA	95320

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	170	100	0	0	0
Number under roof	800	0	0	0	0	0
Maximum number	800	170	100	0	0	0
Average number	800	170	100	0	0	0
Avg live weight (lbs)	1,400	1,450	1,000	0		

Predominant milk cow breed: Holstein

Average milk production: 78 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 24,806.04 *tons per reporting period*

Total nitrogen from manure: 318,639.62 *lbs per reporting period*

After ammonia losses (30% loss applied): 223,047.73 *lbs per reporting period*

Total phosphorus from manure: 52,929.30 *lbs per reporting period*

Total potassium from manure: 157,412.18 *lbs per reporting period*

Total salt from manure: 415,771.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 7,002,900 *gallons*

Total nitrogen generated: 48,311.68 *lbs*

Total phosphorus generated: 6,626.68 *lbs*

Total potassium generated: 60,723.84 *lbs*

Total salt generated: 310,223.21 *lbs*

	9,087,900	*gallons applied*
+	0	*gallons exported*
-	2,085,000	*gallons imported*
=	7,002,900	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal SSJID	Surface water
IW 3	Ground water
IW 4	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

Date	Material type / Description	Quantity	Reporting basis	Moist. (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/05/2023	Corral solids *Corral / Compost from Dairy #2*	3,210.00 *ton*	Dry-weight	54.9	19,900.00	7,280.00	29,300.00		28.40
10/11/2023	Corral solids *Corral Manure from Dairy #3*	2,940.00 *ton*	Dry-weight	53.7	19,100.00	9,630.00	35,300.00		38.00

Date	Material type / Description	Quantity	N (mg/L)	P (mg/L)	K (mg/L)	Salt (mg/L)	TDS (mg/L)
05/01/2023	Process wastewater sludge *Slurry Import*	814,800.00 *gal*	3,910.00	239.00	1,600.00		12,800
05/06/2023	Process wastewater sludge *Slurry*	235,200.00 *gal*	3,910.00	239.00	1,600.00		12,800
06/02/2023	Process wastewater *Wastewater*	1,035,000.00 *gal*	316.00	27.30	319.00		7,180

No commercial or other nutrient imports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	109,617.46	47,295.75	180,937.94	1,856,826.48
Process wastewater	36,989.71	2,329.97	16,774.83	174,171.00
Total imports for all materials	146,607.18	49,625.72	197,712.76	2,030,997.48

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/13/2023	Compost	150.00 *ton*	Dry-weight	29.1		17,700.00	7,760.00	24,900.00		53.20
11/22/2023	Compost	98.00 *ton*	Dry-weight	29.1		17,700.00	7,760.00	24,900.00		53.20

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	6,224.45	2,728.91	8,756.43	187,085.25
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	6,224.45	2,728.91	8,756.43	187,085.25

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Cabral	65	65	2	both	X205-X050-XX25-XXXX
Faria	50	50	2	both	X205-X050-XX10-XXXX
Franzia	12	12	2	manure	X205-X100-XX14-XXXX
Long	6	6	2	process wastewater	X205-X110-XX08-XXXX
Office	7	7	2	process wastewater	X205-X060-XX05-XXXX X205-X060-XX07-XXXX
Rice	21	21	2	process wastewater	X205-X110-XX06-XXXX
Roos	21	21	2	process wastewater	X205-X110-XX08-XXXX
Tony	17	17	2	process wastewater	X205-X110-XX12-XXXX
Zumwalt	22	22	2	process wastewater	X205-X100-XX15-XXXX
Totals for areas that were used for application	221	221	18		
Totals for areas that were not used for application					
Land application area totals	221	221	18		

B. CROPS AND HARVESTS

Cabral

Field name: Cabral

10/31/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 65 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	1,158.30 *ton*	As-is		71.0	5,830.00	1,020.00	8,530.00		11.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	180.00	28.80	149.40	0.00
Total actual harvest content	17.82	207.78	36.35	304.01	1,147.25

Cabral

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 65 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	2,153.77 *ton*	As-is		68.8	4,120.00	580.00	4,010.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	33.13	273.03	38.44	265.74	1,302.60

Faria

Field name: Faria

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 50 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	875.70 *ton*	As-is		68.4	5,150.00	890.00	7,760.00		9.55

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	180.00	28.80	149.40	0.00
Total actual harvest content	17.51	180.39	31.17	271.82	1,057.07

06/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 06/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/03/2023	1,563.97 *ton*	As-is		70.0	3,960.00	570.00	3,950.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.28	247.73	35.66	247.11	1,201.13

Franzia

Field name: Franzia

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 12 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	188.40 *ton*	As-is		54.0	5,840.00	990.00	9,490.00		8.16

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	180.00	28.80	149.40	0.00
Total actual harvest content	15.70	183.38	31.09	297.99	1,178.63

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 12 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/07/2023	364.50 *ton*	As-is		72.6	3,530.00	480.00	4,050.00		7.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.38	214.45	29.16	246.04	1,298.35

Long

Field name: Long

10/30/2002: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 6 Plant date: 10/30/2002

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	111.00 *ton*	As-is		70.2	4,170.00	740.00	6,010.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	180.00	28.80	149.40	0.00
Total actual harvest content	18.50	154.29	27.38	222.37	1,102.60

Long

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 6 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	175.65 *ton*	As-is		64.0	4,610.00	660.00	4,480.00		8.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.28	269.92	38.64	262.30	1,728.40

Office

Field name: Office

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 7 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	129.50 *ton*	As-is		70.2	4,170.00	740.00	6,010.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	180.00	28.80	149.40	0.00
Total actual harvest content	18.50	154.29	27.38	222.37	1,102.60

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 7 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	221.63 *ton*	As-is		64.0	4,610.00	660.00	4,480.00		8.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.66	291.92	41.79	283.69	1,869.29

Rice

Field name: Rice

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 21 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	386.40 *ton*	As-is		68.0	3,390.00	540.00	5,580.00		6.75

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	180.00	28.80	149.40	0.00
Total actual harvest content	18.40	124.75	19.87	205.34	794.88

06/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 21 Plant date: 06/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	646.64 *ton*	As-is		64.2	4,370.00	520.00	4,960.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.79	269.13	32.02	305.46	1,433.08

Roos

Field name: Roos

10/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 21 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	373.80 *ton*	As-is		62.8	3,910.00	780.00	6,710.00		9.48

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	180.00	28.80	149.40	0.00
Total actual harvest content	17.80	139.20	27.77	238.88	1,255.46

Roos

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 21 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	739.89 *ton*	As-is		72.6	3,840.00	540.00	4,610.00		7.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	35.23	270.59	38.05	324.85	1,505.99

Tony

Field name: Tony

10/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 17 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	317.90 *ton*	As-is		71.0	3,420.00	660.00	5,390.00		11.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	180.00	28.80	149.40	0.00
Total actual harvest content	18.70	127.91	24.68	201.59	1,268.98

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	488.15 *ton*	As-is		72.4	3,450.00	530.00	4,620.00		7.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.71	198.13	30.44	265.32	1,188.79

Zumwalt

Field name: Zumwalt

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 22 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	409.20 *ton*	As-is		70.6	3,620.00	670.00	5,950.00		8.28

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	180.00	28.80	149.40	0.00
Total actual harvest content	18.60	134.66	24.92	221.34	905.57

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 22 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/07/2023	724.81 *ton*	As-is		70.2	3,930.00	610.00	3,890.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.95	258.95	40.19	256.32	1,138.87

NUTRIENT BUDGET

A. LAND APPLICATIONS

Cabral - 10/31/2022: Oats, silage-soft dough

Field name: Cabral

Crop: Oats, silage-soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	247.42	85.26	415.63	2,937.00	650.00 *ton*
Application event totals		247.42	85.26	415.63	2,937.00	

Cabral - 06/11/2023: Corn, silage

Field name: Cabral

Crop: Corn, silage

Plant date: 06/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater sludge	358.42	21.91	146.67	1,173.33	714,000.00 *gal*
Application event totals		358.42	21.91	146.67	1,173.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	66.35	8,910,000.00 *gal*
Application event totals		0.00	0.00	0.00	66.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	63.33	8,505,000.00 *gal*
Application event totals		0.00	0.00	0.00	63.33	

Cabral - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	63.33	8,505,000.00 *gal*
Application event totals		0.00	0.00	0.00	63.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	63.33	8,505,000.00 *gal*
Application event totals		0.00	0.00	0.00	63.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	60.32	8,100,000.00 *gal*
Application event totals		0.00	0.00	0.00	60.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	60.32	8,100,000.00 *gal*
Application event totals		0.00	0.00	0.00	60.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	60.32	8,100,000.00 *gal*
Application event totals		0.00	0.00	0.00	60.32	

Faria - 11/01/2022: Oats, silage-soft dough

Field name: Faria

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Faria - 11/01/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	242.47	83.56	407.32	2,878.26	490.00 *ton*
Application event totals		242.47	83.56	407.32	2,878.26	

Faria - 06/09/2023: Corn, silage

Field name: Faria

Crop: Corn, silage

Plant date: 06/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater sludge	328.90	20.10	134.59	1,076.71	504,000.00 *gal*
Application event totals		328.90	20.10	134.59	1,076.71	

05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	66.65	6,885,000.00 *gal*
Application event totals		0.00	0.00	0.00	66.65	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	62.73	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	62.73	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	62.73	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	62.73	

Faria - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	62.73	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	62.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	62.73	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	62.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	60.77	6,277,500.00 *gal*
Application event totals		0.00	0.00	0.00	60.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	58.81	6,075,000.00 *gal*
Application event totals		0.00	0.00	0.00	58.81	

Franzia - 11/02/2022: Oats, silage-soft dough

Field name: Franzia

Crop: Oats, silage-soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	247.42	85.26	415.63	2,937.00	120.00 *ton*
Application event totals		247.42	85.26	415.63	2,937.00	

Franzia - 06/12/2023: Corn, silage

Field name: Franzia

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	237.31	99.07	264.96	2,979.84	120.00 *ton*
Application event totals		237.31	99.07	264.96	2,979.84	

05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	65.34	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.34	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	65.34	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.34	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	65.34	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.34	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	65.34	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.34	

08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	61.26	1,518,750.00 *gal*
Application event totals		0.00	0.00	0.00	61.26	

Franzia - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	61.26	1,518,750.00 *gal*
Application event totals		0.00	0.00	0.00	61.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	57.17	1,417,500.00 *gal*
Application event totals		0.00	0.00	0.00	57.17	

Long - 10/30/2002: Oats, silage-soft dough

Field name: Long

Crop: Oats, silage-soft dough

Plant date: 10/30/2002

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	208.19	43.79	337.90	2,354.54	297,000.00 *gal*
Application event totals		208.19	43.79	337.90	2,354.54	

Long - 06/10/2023: Corn, silage

Field name: Long

Crop: Corn, silage

Plant date: 06/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Long - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	114.19	14.03	123.74	1,274.11	132,000.00 *gal*
Canal SSJID	Surface water	0.00	0.00	0.00	65.34	810,000.00 *gal*
Application event totals		114.19	14.03	123.74	1,339.46	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	65.34	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.34	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	94.94	11.07	168.67	899.59	115,500.00 *gal*
Canal SSJID	Surface water	0.00	0.00	0.00	57.17	708,750.00 *gal*
Application event totals		94.94	11.07	168.67	956.76	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	65.34	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.34	

08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	94.94	11.07	168.67	899.59	115,500.00 *gal*
Canal SSJID	Surface water	0.00	0.00	0.00	57.17	708,750.00 *gal*
Application event totals		94.94	11.07	168.67	956.76	

Long - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	65.34	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	57.17	708,750.00 *gal*
Application event totals		0.00	0.00	0.00	57.17	

Office - 10/29/2022: Oats, silage-soft dough

Field name: Office

Crop: Oats, silage-soft dough Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater sludge	207.49	49.77	385.54	0.00	294,000.00 *gal*
Application event totals		207.49	49.77	385.54	0.00	

Office - 06/10/2023: Corn, silage

Field name: Office

Crop: Corn, silage Plant date: 06/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Office - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater sludge	352.39	21.54	144.20	1,153.61	75,600.00 *gal*
Application event totals		352.39	21.54	144.20	1,153.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	70.01	1,012,500.00 *gal*
Application event totals		0.00	0.00	0.00	70.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	63.01	911,250.00 *gal*
Application event totals		0.00	0.00	0.00	63.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	63.01	911,250.00 *gal*
Application event totals		0.00	0.00	0.00	63.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	63.01	911,250.00 *gal*
Application event totals		0.00	0.00	0.00	63.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	63.01	911,250.00 *gal*
Application event totals		0.00	0.00	0.00	63.01	

Office - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	63.01	911,250.00 *gal*
Application event totals		0.00	0.00	0.00	63.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	56.01	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	56.01	

Rice - 10/29/2022: Oats, silage-soft dough

Field name: Rice

Crop: Oats, silage-soft dough Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	171.84	36.14	278.90	1,943.43	858,000.00 *gal*
Application event totals		171.84	36.14	278.90	1,943.43	

Rice - 06/09/2023: Corn, silage

Field name: Rice

Crop: Corn, silage Plant date: 06/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Rice - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	114.19	14.03	123.74	1,274.11	462,000.00 *gal*
Canal SSJID	Surface water	0.00	0.00	0.00	65.34	2,835,000.00 *gal*
Application event totals		114.19	14.03	123.74	1,339.46	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	65.34	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.34	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	100.75	11.75	179.00	954.67	429,000.00 *gal*
Canal SSJID	Surface water	0.00	0.00	0.00	60.67	2,632,500.00 *gal*
Application event totals		100.75	11.75	179.00	1,015.34	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	60.67	2,632,500.00 *gal*
Application event totals		0.00	0.00	0.00	60.67	

08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	93.00	10.84	165.23	881.23	396,000.00 *gal*
Canal SSJID	Surface water	0.00	0.00	0.00	56.01	2,430,000.00 *gal*
Application event totals		93.00	10.84	165.23	937.24	

Rice - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	60.67	2,632,500.00 *gal*
Application event totals		0.00	0.00	0.00	60.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	60.67	2,632,500.00 *gal*
Application event totals		0.00	0.00	0.00	60.67	

Roos - 10/30/2022: Oats, silage-soft dough

Field name: Roos

Crop: Oats, silage-soft dough Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	185.06	38.92	300.35	2,092.93	924,000.00 *gal*
Application event totals		185.06	38.92	300.35	2,092.93	

Roos - 06/10/2023: Corn, silage

Field name: Roos

Crop: Corn, silage Plant date: 06/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Roos - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	106.04	13.02	114.90	1,183.11	429,000.00 *gal*
Canal SSJID	Surface water	0.00	0.00	0.00	60.67	2,632,500.00 *gal*
Application event totals		106.04	13.02	114.90	1,243.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	65.34	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	96.88	11.29	172.12	917.95	412,500.00 *gal*
Canal SSJID	Surface water	0.00	0.00	0.00	58.34	2,531,250.00 *gal*
Application event totals		96.88	11.29	172.12	976.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	65.34	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	93.00	10.84	165.23	881.23	396,000.00 *gal*
Canal SSJID	Surface water	0.00	0.00	0.00	56.01	2,430,000.00 *gal*
Application event totals		93.00	10.84	165.23	937.24	

Roos - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	60.67	2,632,500.00 *gal*
Application event totals		0.00	0.00	0.00	60.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	60.67	2,632,500.00 *gal*
Application event totals		0.00	0.00	0.00	60.67	

Tony - 10/30/2022: Oats, silage-soft dough

Field name: Tony

Crop: Oats, silage-soft dough Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	163.29	34.34	265.02	1,846.70	660,000.00 *gal*
Application event totals		163.29	34.34	265.02	1,846.70	

Tony - 06/10/2023: Corn, silage

Field name: Tony

Crop: Corn, silage Plant date: 06/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Tony - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	100.76	12.38	109.18	1,124.22	330,000.00 *gal*
Canal SSJID	Surface water	0.00	0.00	0.00	57.65	2,025,000.00 *gal*
Application event totals		100.76	12.38	109.18	1,181.87	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	63.42	2,227,500.00 *gal*
Application event totals		0.00	0.00	0.00	63.42	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	86.16	10.05	153.08	816.44	297,000.00 *gal*
Canal SSJID	Surface water	0.00	0.00	0.00	51.89	1,822,500.00 *gal*
Application event totals		86.16	10.05	153.08	868.32	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	63.42	2,227,500.00 *gal*
Application event totals		0.00	0.00	0.00	63.42	

08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	76.59	8.93	136.07	725.72	264,000.00 *gal*
Canal SSJID	Surface water	0.00	0.00	0.00	46.12	1,620,000.00 *gal*
Application event totals		76.59	8.93	136.07	771.84	

Tony - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	63.42	2,227,500.00 *gal*
Application event totals		0.00	0.00	0.00	63.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	57.65	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.65	

Zumwalt - 11/02/2022: Oats, silage-soft dough

Field name: Zumwalt

Crop: Oats, silage-soft dough Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater sludge	169.76	40.72	315.44	0.00	756,000.00 *gal*
Application event totals		169.76	40.72	315.44	0.00	

Zumwalt - 06/12/2023: Corn, silage

Field name: Zumwalt

Crop: Corn, silage Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Zumwalt - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater sludge	336.37	20.56	137.65	1,101.18	226,800.00 *gal*
Application event totals		336.37	20.56	137.65	1,101.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	66.83	3,037,500.00 *gal*
Application event totals		0.00	0.00	0.00	66.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	66.83	3,037,500.00 *gal*
Application event totals		0.00	0.00	0.00	66.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	66.83	3,037,500.00 *gal*
Application event totals		0.00	0.00	0.00	66.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	62.37	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	62.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	62.37	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	62.37	

Zumwalt - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	62.37	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	62.37	

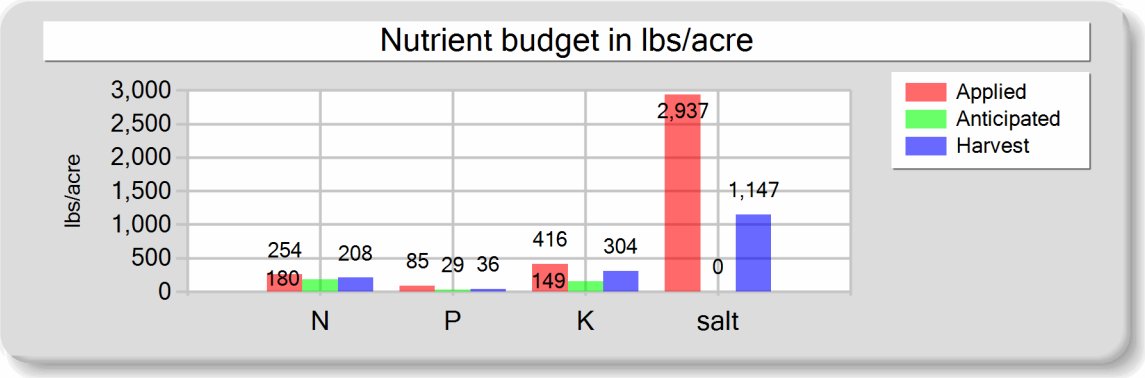
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSJID	Surface water	0.00	0.00	0.00	57.92	2,632,500.00 *gal*
Application event totals		0.00	0.00	0.00	57.92	

B. NUTRIENT BUDGET

Cabral - 10/31/2022: Oats, silage-soft dough

Field name: Cabral | Crop: Oats, silage-soft dough | Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	247.42	85.26	415.63	2,937.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	254.42	85.26	415.63	2,937.00
Anticipated crop nutrient removal	180.00	28.80	149.40	0.00
Actual crop nutrient removal	207.78	36.35	304.01	1,147.25
Nutrient balance	46.64	48.91	111.62	1,789.75
Applied to removed ratio	1.22	2.35	1.37	2.56

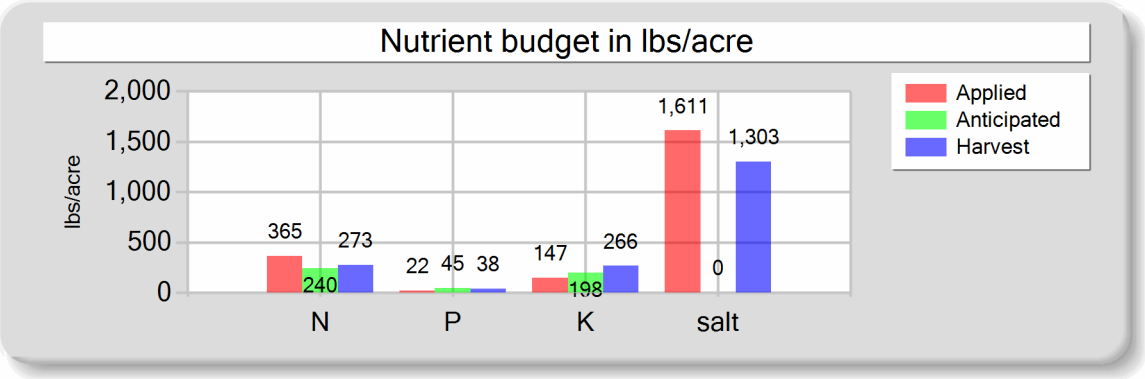
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Cabral - 06/11/2023: Corn, silage

Field name: Cabral | Crop: Corn, silage | Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	358.42	21.91	146.67	1,173.33
Fresh water	0.00	0.00	0.00	437.28
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	365.42	21.91	146.67	1,610.62
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	273.03	38.44	265.74	1,302.60
Nutrient balance	92.38	-16.53	-119.08	308.02
Applied to removed ratio	1.34	0.57	0.55	1.24

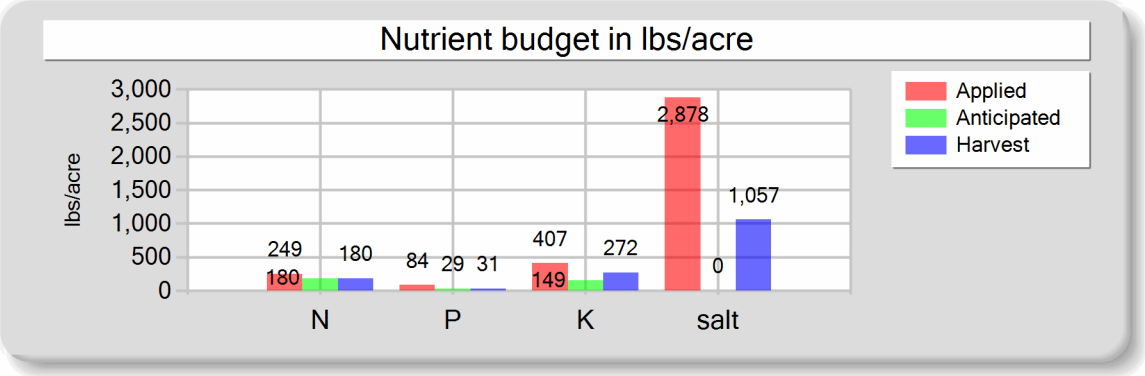
Fresh water applied	
58,725,000.00	*gallons*
2,162.64	*acre-inches*
33.27	*inches/acre*

Process wastewater applied	
714,000.00	*gallons*
26.29	*acre-inches*
0.40	*inches/acre*

Total harvests for the crop	
1	*harvests*

Faria - 11/01/2022: Oats, silage-soft dough

Field name: Faria | Crop: Oats, silage-soft dough | Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	242.47	83.56	407.32	2,878.26
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	249.47	83.56	407.32	2,878.26
Anticipated crop nutrient removal	180.00	28.80	149.40	0.00
Actual crop nutrient removal	180.39	31.17	271.82	1,057.07
Nutrient balance	69.08	52.38	135.50	1,821.19
Applied to removed ratio	1.38	2.68	1.50	2.72

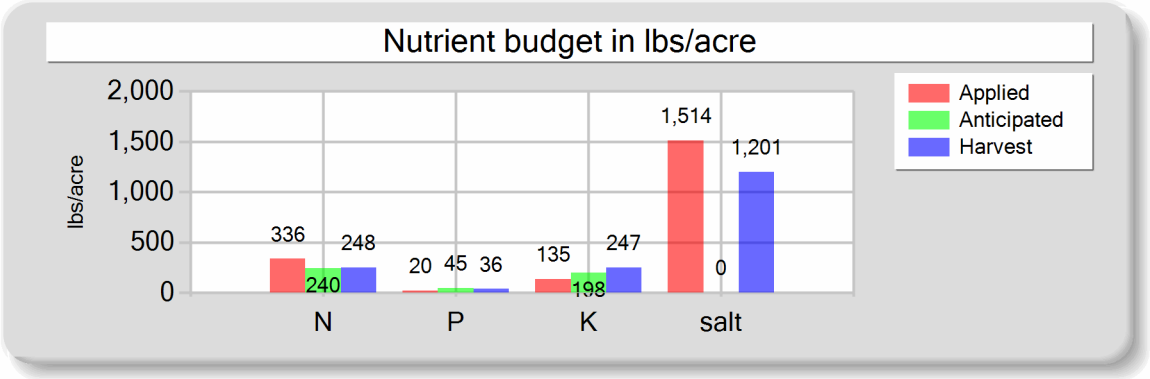
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Faria - 06/09/2023: Corn, silage

Field name: Faria | Crop: Corn, silage | Plant date: 06/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	328.90	20.10	134.59	1,076.71
Fresh water	0.00	0.00	0.00	437.13
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	335.90	20.10	134.59	1,513.84
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	247.73	35.66	247.11	1,201.13
Nutrient balance	88.17	-15.55	-112.52	312.71
Applied to removed ratio	1.36	0.56	0.54	1.26

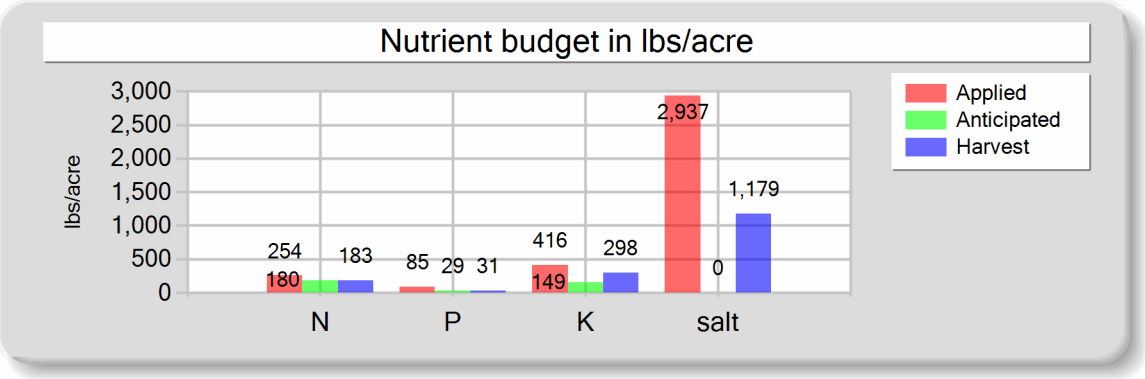
Fresh water applied	
45,157,500.00	*gallons*
1,663.00	*acre-inches*
33.26	*inches/acre*

Process wastewater applied	
504,000.00	*gallons*
18.56	*acre-inches*
0.37	*inches/acre*

Total harvests for the crop	
1	*harvests*

Franzia - 11/02/2022: Oats, silage-soft dough

Field name: Franzia | Crop: Oats, silage-soft dough | Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	247.42	85.26	415.63	2,937.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	254.42	85.26	415.63	2,937.00
Anticipated crop nutrient removal	180.00	28.80	149.40	0.00
Actual crop nutrient removal	183.38	31.09	297.99	1,178.63
Nutrient balance	71.04	54.18	117.64	1,758.37
Applied to removed ratio	1.39	2.74	1.39	2.49

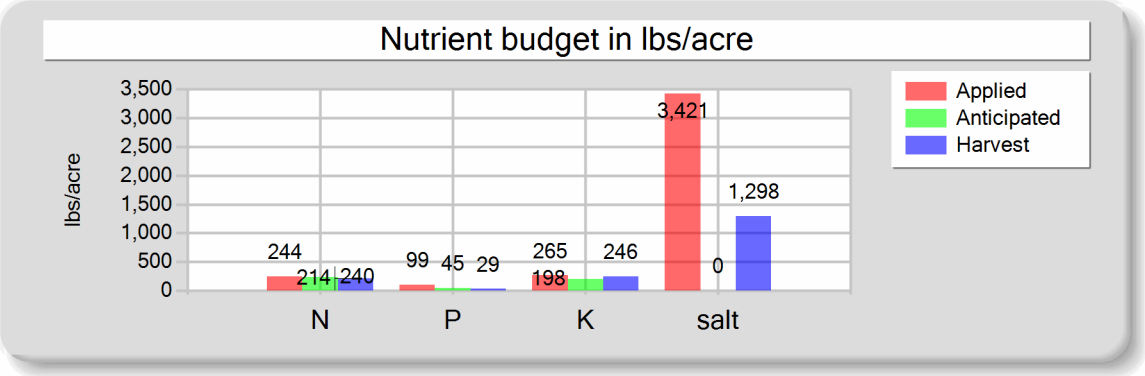
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Franzia - 06/12/2023: Corn, silage

Field name: Franzia | Crop: Corn, silage | Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	237.31	99.07	264.96	2,979.84
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	441.05
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	244.31	99.07	264.96	3,420.89
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	214.45	29.16	246.04	1,298.35
Nutrient balance	29.86	69.91	18.92	2,122.55
Applied to removed ratio	1.14	3.40	1.08	2.63

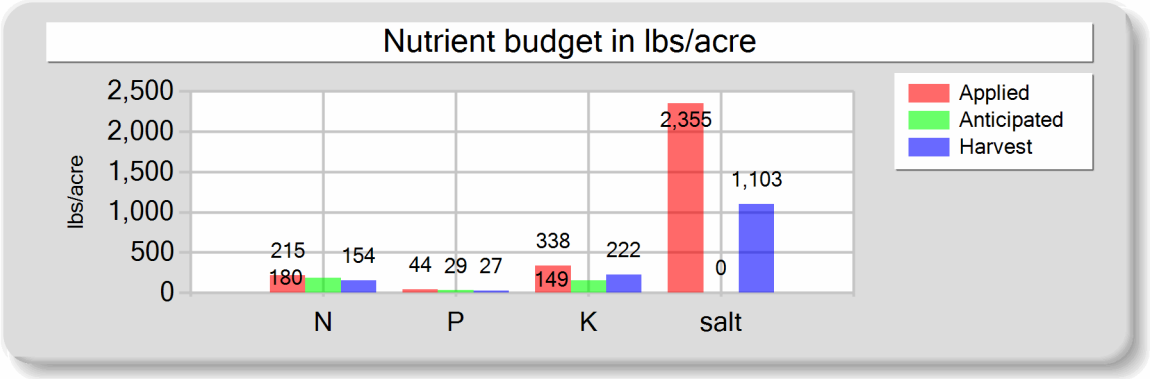
Fresh water applied	
10,935,000.00	*gallons*
402.70	*acre-inches*
33.56	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Long - 10/30/2002: Oats, silage-soft dough

Field name: Long | Crop: Oats, silage-soft dough | Plant date: 10/30/2002



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	208.19	43.79	337.90	2,354.54
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	215.19	43.79	337.90	2,354.54
Anticipated crop nutrient removal	180.00	28.80	149.40	0.00
Actual crop nutrient removal	154.29	27.38	222.37	1,102.60
Nutrient balance	60.90	16.41	115.53	1,251.94
Applied to removed ratio	1.39	1.60	1.52	2.14

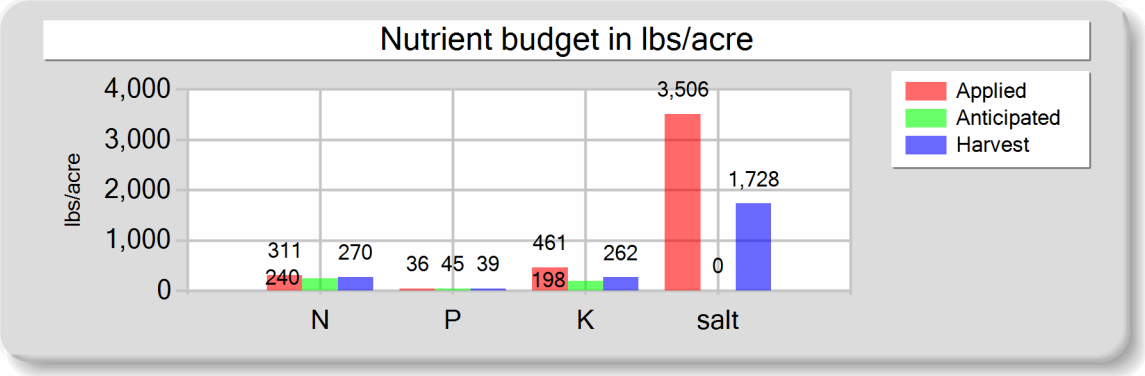
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
297,000.00	*gallons*
10.94	*acre-inches*
1.82	*inches/acre*

Total harvests for the crop	
1	*harvests*

Long - 06/10/2023: Corn, silage

Field name: Long | Crop: Corn, silage | Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	304.07	36.16	461.09	3,073.30
Fresh water	0.00	0.00	0.00	432.89
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	311.07	36.16	461.09	3,506.18
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	269.92	38.64	262.30	1,728.40
Nutrient balance	41.16	-2.48	198.78	1,777.79
Applied to removed ratio	1.15	0.94	1.76	2.03

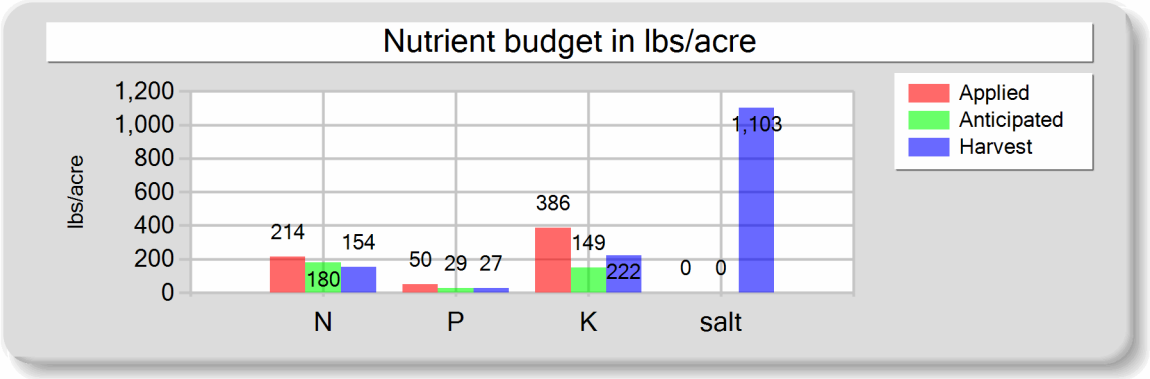
Fresh water applied	
5,366,250.00	*gallons*
197.62	*acre-inches*
32.94	*inches/acre*

Process wastewater applied	
363,000.00	*gallons*
13.37	*acre-inches*
2.23	*inches/acre*

Total harvests for the crop	
1	*harvests*

Office - 10/29/2022: Oats, silage-soft dough

Field name: Office | Crop: Oats, silage-soft dough | Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	207.49	49.77	385.54	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	214.49	49.77	385.54	0.00
Anticipated crop nutrient removal	180.00	28.80	149.40	0.00
Actual crop nutrient removal	154.29	27.38	222.37	1,102.60
Nutrient balance	60.20	22.39	163.17	-1,102.60
Applied to removed ratio	1.39	1.82	1.73	0.00

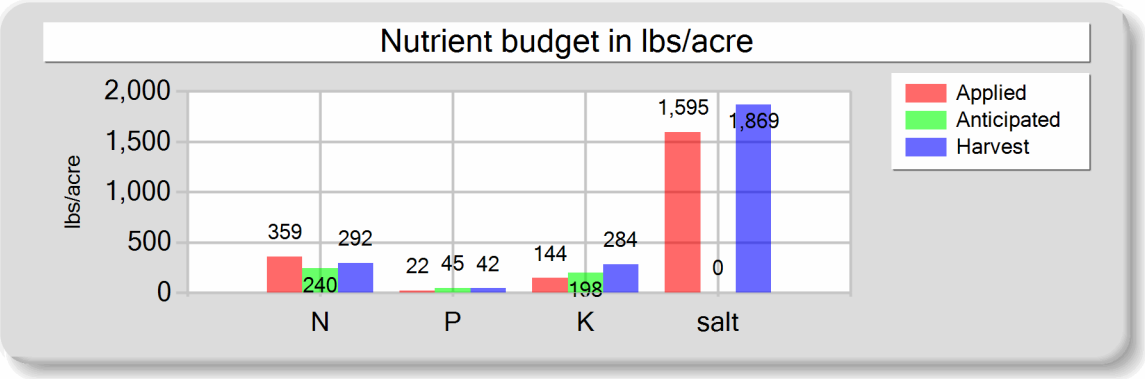
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
294,000.00	*gallons*
10.83	*acre-inches*
1.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

Office - 06/10/2023: Corn, silage

Field name: Office | Crop: Corn, silage | Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	352.39	21.54	144.20	1,153.61
Fresh water	0.00	0.00	0.00	441.05
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	359.39	21.54	144.20	1,594.67
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	291.92	41.79	283.69	1,869.29
Nutrient balance	67.47	-20.25	-139.48	-274.62
Applied to removed ratio	1.23	0.52	0.51	0.85

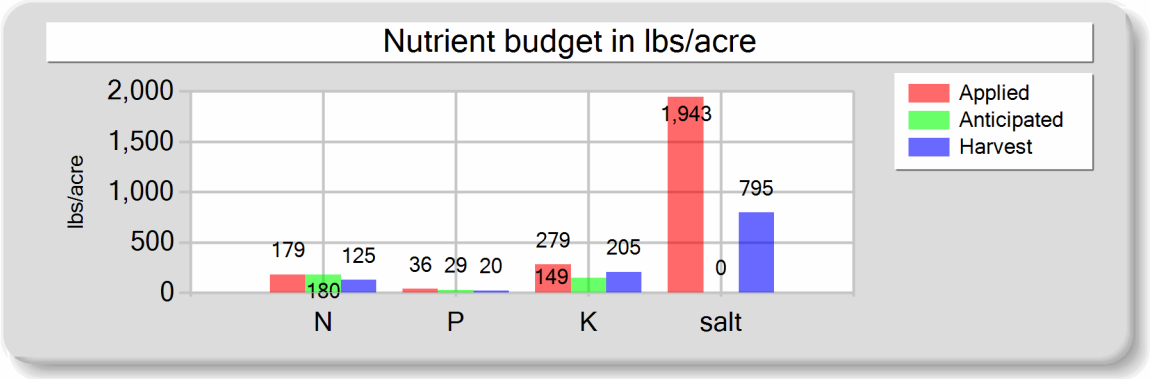
Fresh water applied	
6,378,750.00	*gallons*
234.91	*acre-inches*
33.56	*inches/acre*

Process wastewater applied	
75,600.00	*gallons*
2.78	*acre-inches*
0.40	*inches/acre*

Total harvests for the crop	
1	*harvests*

Rice - 10/29/2022: Oats, silage-soft dough

Field name: Rice | Crop: Oats, silage-soft dough | Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	171.84	36.14	278.90	1,943.43
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	178.84	36.14	278.90	1,943.43
Anticipated crop nutrient removal	180.00	28.80	149.40	0.00
Actual crop nutrient removal	124.75	19.87	205.34	794.88
Nutrient balance	54.09	16.27	73.56	1,148.55
Applied to removed ratio	1.43	1.82	1.36	2.44

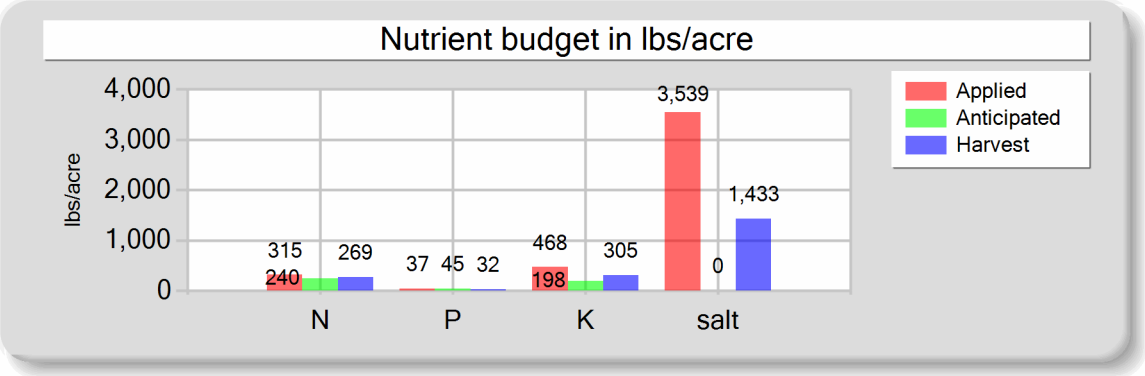
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
858,000.00	*gallons*
31.60	*acre-inches*
1.50	*inches/acre*

Total harvests for the crop	
1	*harvests*

Rice - 06/09/2023: Corn, silage

Field name: Rice | Crop: Corn, silage | Plant date: 06/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	307.95	36.61	467.97	3,110.01
Fresh water	0.00	0.00	0.00	429.39
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	314.95	36.61	467.97	3,539.40
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	269.13	32.02	305.46	1,433.08
Nutrient balance	45.82	4.59	162.51	2,106.32
Applied to removed ratio	1.17	1.14	1.53	2.47

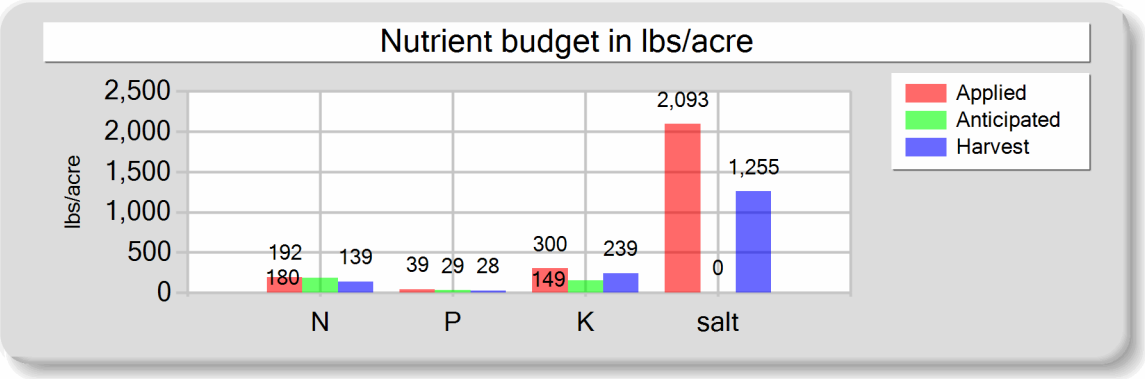
Fresh water applied	
18,630,000.00	*gallons*
686.08	*acre-inches*
32.67	*inches/acre*

Process wastewater applied	
1,287,000.00	*gallons*
47.40	*acre-inches*
2.26	*inches/acre*

Total harvests for the crop	
1	*harvests*

Roos - 10/30/2022: Oats, silage-soft dough

Field name: Roos | Crop: Oats, silage-soft dough | Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	185.06	38.92	300.35	2,092.93
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	192.06	38.92	300.35	2,092.93
Anticipated crop nutrient removal	180.00	28.80	149.40	0.00
Actual crop nutrient removal	139.20	27.77	238.88	1,255.46
Nutrient balance	52.86	11.15	61.48	837.47
Applied to removed ratio	1.38	1.40	1.26	1.67

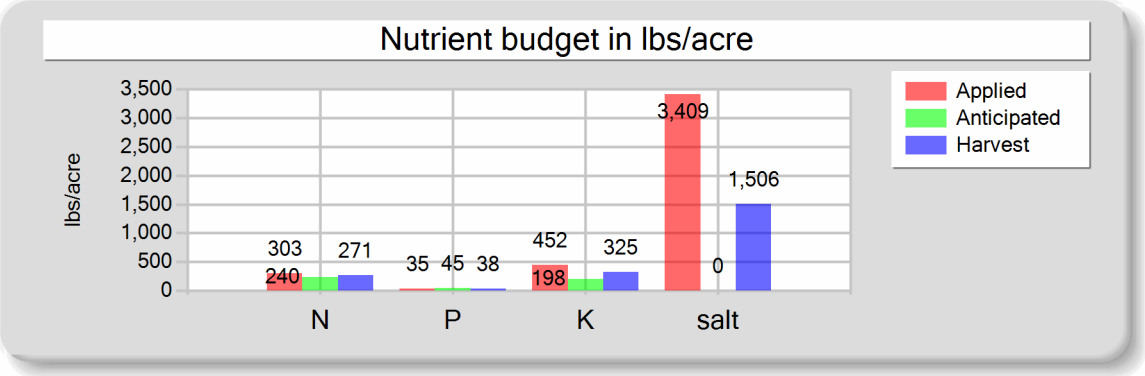
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
924,000.00	*gallons*
34.03	*acre-inches*
1.62	*inches/acre*

Total harvests for the crop	
1	*harvests*

Roos - 06/10/2023: Corn, silage

Field name: Roos | Crop: Corn, silage | Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	295.91	35.16	452.25	2,982.29
Fresh water	0.00	0.00	0.00	427.05
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	302.91	35.16	452.25	3,409.34
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	270.59	38.05	324.85	1,505.99
Nutrient balance	32.33	-2.89	127.40	1,903.35
Applied to removed ratio	1.12	0.92	1.39	2.26

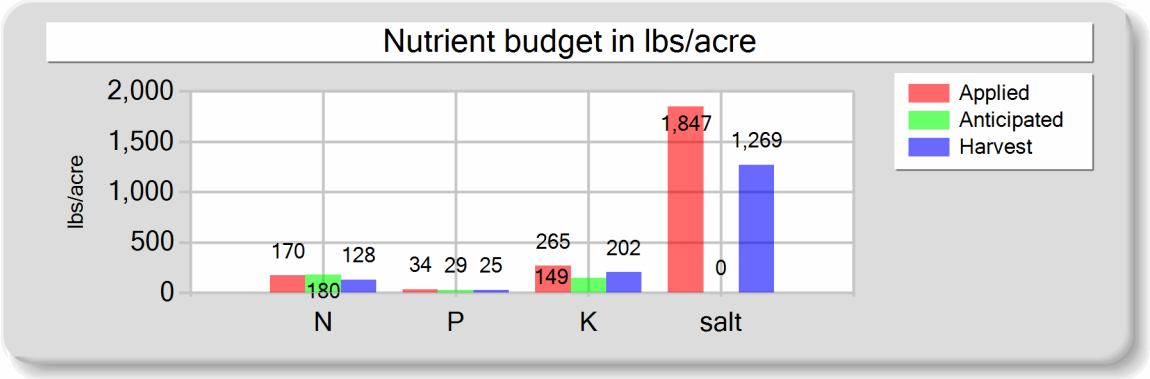
Fresh water applied	
18,528,750.00	*gallons*
682.35	*acre-inches*
32.49	*inches/acre*

Process wastewater applied	
1,237,500.00	*gallons*
45.57	*acre-inches*
2.17	*inches/acre*

Total harvests for the crop	
1	*harvests*

Tony - 10/30/2022: Oats, silage-soft dough

Field name: Tony | Crop: Oats, silage-soft dough | Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	163.29	34.34	265.02	1,846.70
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	170.29	34.34	265.02	1,846.70
Anticipated crop nutrient removal	180.00	28.80	149.40	0.00
Actual crop nutrient removal	127.91	24.68	201.59	1,268.98
Nutrient balance	42.38	9.66	63.43	577.72
Applied to removed ratio	1.33	1.39	1.31	1.46

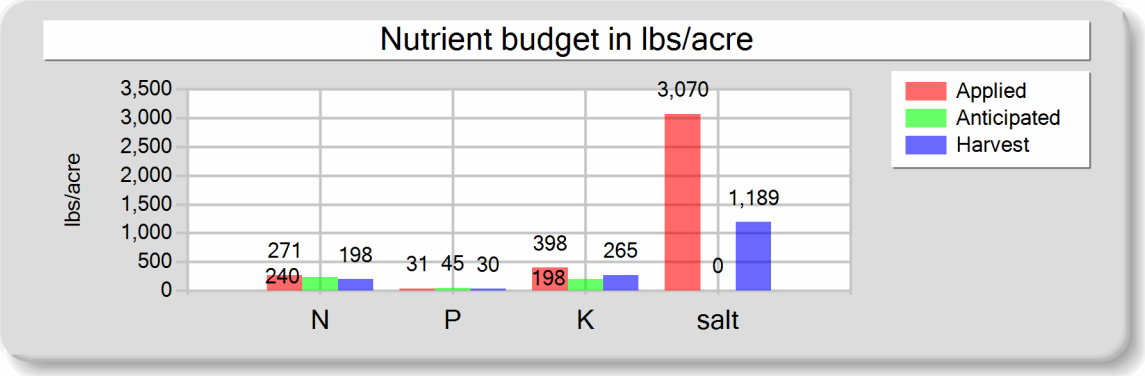
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
660,000.00	*gallons*
24.31	*acre-inches*
1.43	*inches/acre*

Total harvests for the crop	
1	*harvests*

Tony - 06/10/2023: Corn, silage

Field name: Tony | Crop: Corn, silage | Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	263.51	31.35	398.34	2,666.37
Fresh water	0.00	0.00	0.00	403.58
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	270.51	31.35	398.34	3,069.95
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	198.13	30.44	265.32	1,188.79
Nutrient balance	72.38	0.91	133.01	1,881.16
Applied to removed ratio	1.37	1.03	1.50	2.58

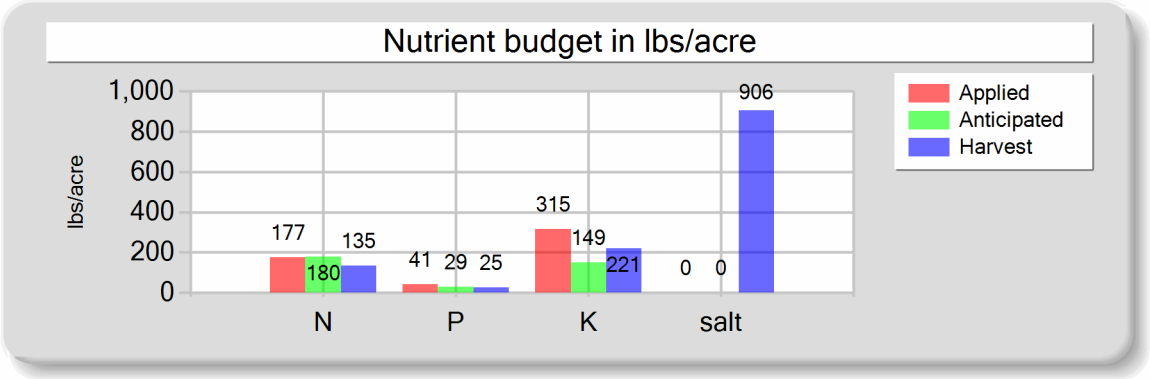
Fresh water applied	
14,175,000.00	*gallons*
522.02	*acre-inches*
30.71	*inches/acre*

Process wastewater applied	
891,000.00	*gallons*
32.81	*acre-inches*
1.93	*inches/acre*

Total harvests for the crop	
1	*harvests*

Zumwalt - 11/02/2022: Oats, silage-soft dough

Field name: Zumwalt | Crop: Oats, silage-soft dough | Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	169.76	40.72	315.44	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	176.76	40.72	315.44	0.00
Anticipated crop nutrient removal	180.00	28.80	149.40	0.00
Actual crop nutrient removal	134.66	24.92	221.34	905.57
Nutrient balance	42.10	15.80	94.10	-905.57
Applied to removed ratio	1.31	1.63	1.43	0.00

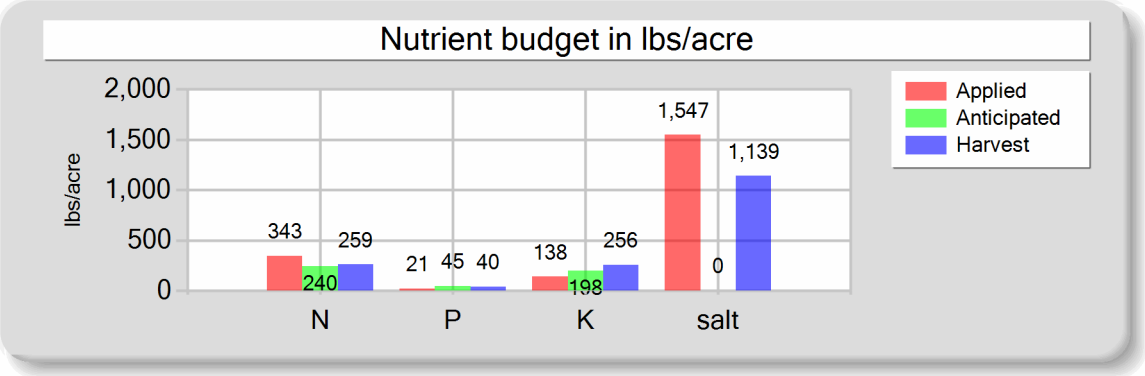
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
756,000.00	*gallons*
27.84	*acre-inches*
1.27	*inches/acre*

Total harvests for the crop	
1	*harvests*

Zumwalt - 06/12/2023: Corn, silage

Field name: Zumwalt Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	336.37	20.56	137.65	1,101.18
Fresh water	0.00	0.00	0.00	445.51
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	343.37	20.56	137.65	1,546.69
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	258.95	40.19	256.32	1,138.87
Nutrient balance	84.42	-19.63	-118.67	407.81
Applied to removed ratio	1.33	0.51	0.54	1.36

Fresh water applied	
20,250,000.00	*gallons*
745.74	*acre-inches*
33.90	*inches/acre*

Process wastewater applied	
226,800.00	*gallons*
8.35	*acre-inches*
0.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 09/12/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 55.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,800.00	9,580.00	46,700.00	27,200.00	14,000.00	12,900.00	9,600.00	2,290.00		33.00
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Compost Dairy 1

Sample and source description: Compost Dairy 1

Sample date: 04/17/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 13.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,000.00	9,000.00	17,400.00							55.90
DL	100.00	100.00	100.00							0.01

Corral Solids Dairy 1

Sample and source description: Corral Solids Dairy 1

Sample date: 04/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	30,900.00	12,900.00	34,500.00	37,200.00	16,500.00	10,800.00	5,800.00	2,590.00		38.80
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Compost - Dairy 1

Sample and source description: Compost - Dairy 1

Sample date: 10/09/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 29.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,700.00	7,760.00	24,900.00							53.20
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/09/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 54.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,900.00	7,280.00	29,300.00							28.40
DL	100.00	100.00	100.00							0.01

Corral Solids Dairy 1

Sample and source description: Corral Solids Dairy 1

Sample date: 10/09/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,600.00	1,970.00	27,200.00							27.80
DL	100.00	100.00	100.00							0.01

Corral Solids Dairy 3

Sample and source description: Corral Solids Dairy 3

Sample date: 10/09/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,100.00	9,630.00	35,300.00							38.00
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

Slurry

Sample and source description: Slurry

Sample date: 10/24/2022 Material type: Process wastewater sludge Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	592.00	592.00	0.00	0.00	142.00	1,100.00								7,510.00	0
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 02/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	504.00	311.00	0.00	0.00	106.00	818.00	266.00	154.00	246.00	3,180.00	0.00	350.00	509.00	7,100.00	5,700
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	50.00	1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 04/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	622.00	342.00	0.00	0.00	76.40	674.00								1,880.00	6,940
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ3 - Imported From #3

Sample and source description: WWQ3 - Imported From #3

Sample date: 04/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	316.00	140.00	0.00	0.00	27.30	319.00								5,990.00	7,180
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Slurry

Sample and source description: Slurry

Sample date: 05/01/2023 Material type: Process wastewater sludge Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3,910.00	1,270.00	0.00	0.00	239.00	1,600.00								17,500.00	12,800
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 07/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	591.00	454.00	0.00	0.00	68.90	1,050.00								8,560.00	5,600
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Slurry

Sample and source description: Slurry

Sample date: 10/09/2023 Material type: Process wastewater sludge Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	484.00	64.40	0.00	0.00	104.00	754.00								5,470.00	2,560
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	728.00	442.00	0.00	0.00	62.10	954.00								7,780.00	5,360
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Canal SSJID

Canal SSJID

Sample description: Canal SSJID

Sample date: 08/18/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								89.20	58
DL	1.00	0.50	0.50								10.00	10

IW 3

IW 3

IW 3

Sample description: IW 3

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	19.80	0.00	19.50								801.00	490
DL	1.00	0.50	0.10								10.00	10

IW 4

IW 4

Sample description: IW 4

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	24.20	0.00	23.90								776.00	510
DL	1.00	0.50	0.50								10.00	10

D. SOIL ANALYSES

Cabral

Soil

Sample and source description: Soil

Sample date: 11/06/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			100.00				
DL			4.00				

Faria

Soil

Sample and source description: Soil

Sample date: 11/06/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			80.00				
DL			4.00				

Franzia

foil

Sample and source description: foil

Sample date: 11/06/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			63.00				
DL			4.00				

Long

Soil

Sample and source description: Soil

Sample date: 11/06/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			82.00				
DL			4.00				

Office

Office

Soil

Sample and source description: Soil

Sample date: 11/06/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			82.00				
DL			4.00				

Rice

Soil

Sample and source description: Soil

Sample date: 11/06/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			96.00				
DL			4.00				

Roos

Soil

Sample and source description: Soil

Sample date: 11/06/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			112.00				
DL			4.00				

Tony

Tony

Soil

Sample and source description: Soil

Sample date: 11/06/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			112.00				
DL			4.00				

Zumwalt

Soil

Sample and source description: Soil

Sample date: 11/06/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			92.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

Cabral - 10/31/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,830.00	1,020.00	8,530.00		11.10
DL	100.00	100.00	100.00		0.01

Cabral - 06/11/2023: Corn, silage

Cabral - 06/11/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,120.00	580.00	4,010.00		6.30
DL	100.00	100.00	100.00		0.01

Faria - 11/01/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,150.00	890.00	7,760.00		9.55
DL	100.00	100.00	100.00		0.01

Faria - 06/09/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,960.00	570.00	3,950.00		6.40
DL	100.00	100.00	100.00		0.01

Franzia - 06/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,530.00	480.00	4,050.00		7.80
DL	100.00	100.00	100.00		0.01

Long - 10/30/2002: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,170.00	740.00	6,010.00		10.00
DL	100.00	100.00	100.00		0.01

Long - 06/10/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,610.00	660.00	4,480.00		8.20
DL	100.00	100.00	100.00		0.01

Office - 10/29/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,170.00	740.00	6,010.00		10.00
DL	100.00	100.00	100.00		0.01

Office - 06/10/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,610.00	660.00	4,480.00		8.20
DL	100.00	100.00	100.00		0.01

Rice - 10/29/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,390.00	540.00	5,580.00		6.75
DL	100.00	100.00	100.00		0.01

Rice - 06/09/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,370.00	520.00	4,960.00		6.50
DL	100.00	100.00	100.00		0.01

Roos - 10/30/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,910.00	780.00	6,710.00		9.48
DL	100.00	100.00	100.00		0.01

Roos - 06/10/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,840.00	540.00	4,610.00		7.80
DL	100.00	100.00	100.00		0.01

Tony - 10/30/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,420.00	660.00	5,390.00		11.70
DL	100.00	100.00	100.00		0.01

Tony - 06/10/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,450.00	530.00	4,620.00		7.50
DL	100.00	100.00	100.00		0.01

Zumwalt - 11/02/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,620.00	670.00	5,950.00		8.28
DL	100.00	100.00	100.00		0.01

Zumwalt - 06/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,930.00	610.00	3,890.00		5.80
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

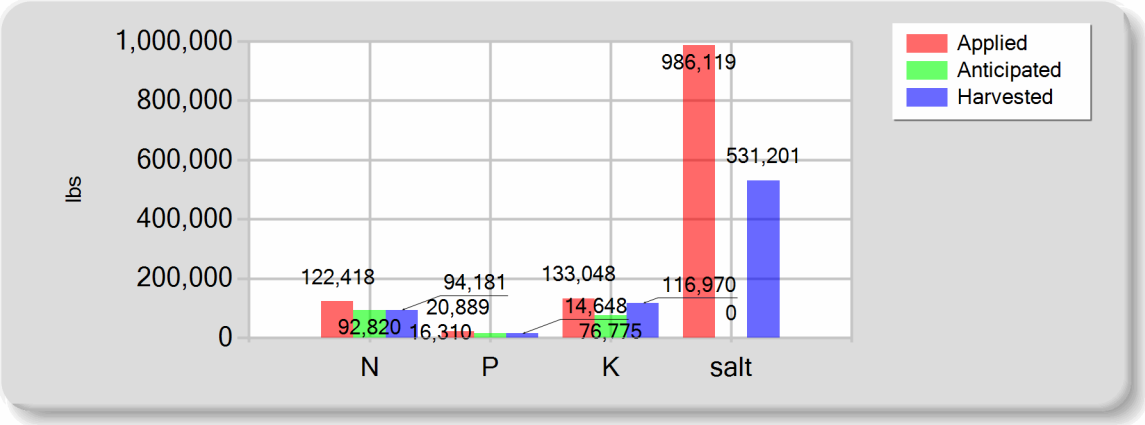
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

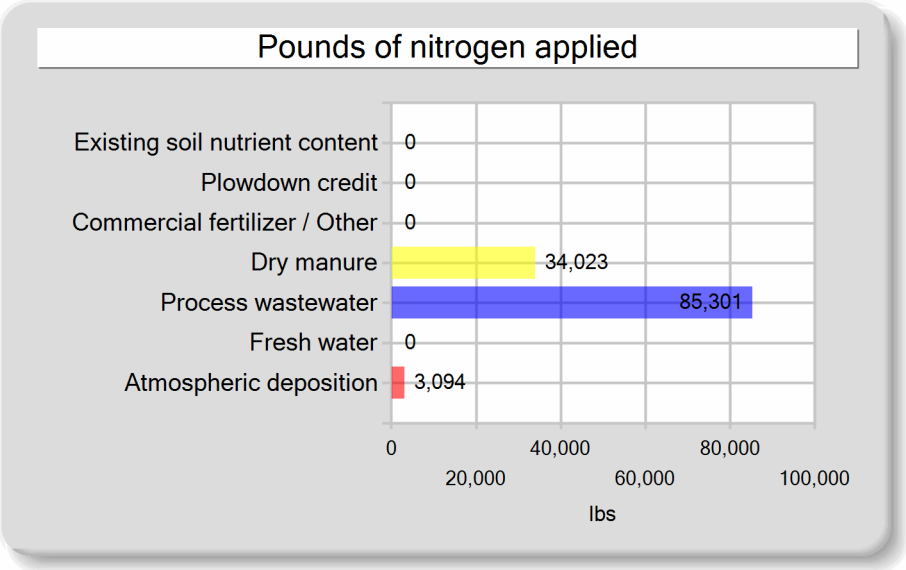
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

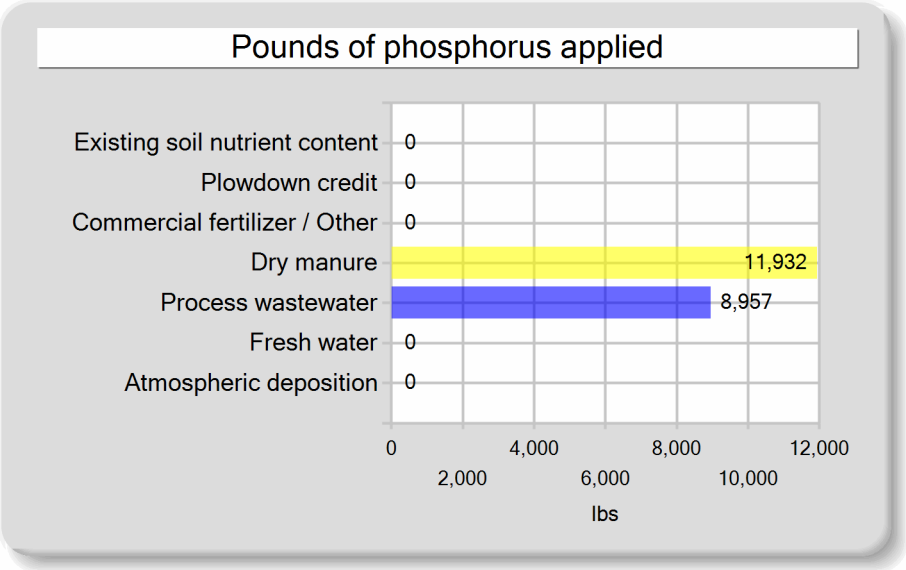
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	34,022.66	11,931.88	55,548.90	405,820.08
Process wastewater	85,301.39	8,956.65	77,498.67	484,394.20
Fresh water	0.00	0.00	0.00	95,904.77
Atmospheric deposition	3,094.00	0.00	0.00	0.00
Total nutrients applied	122,418.06	20,888.53	133,047.57	986,119.05
Anticipated crop nutrient removal	92,820.00	16,309.80	76,775.40	0.00
Actual crop nutrient removal	94,180.95	14,647.97	116,969.83	531,200.86
Nutrient balance	28,237.11	6,240.56	16,077.75	454,918.19
Applied to removed ratio	1.30	1.43	1.14	1.86

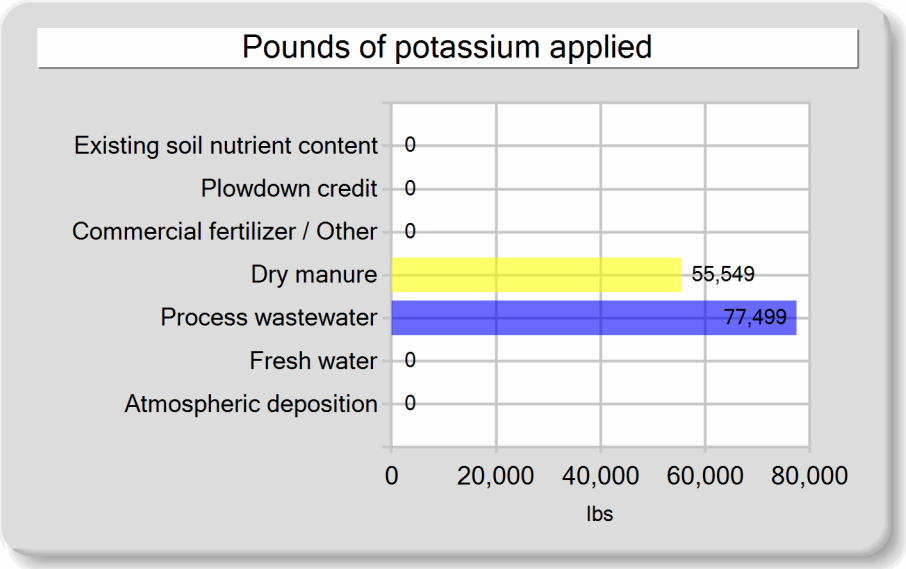
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

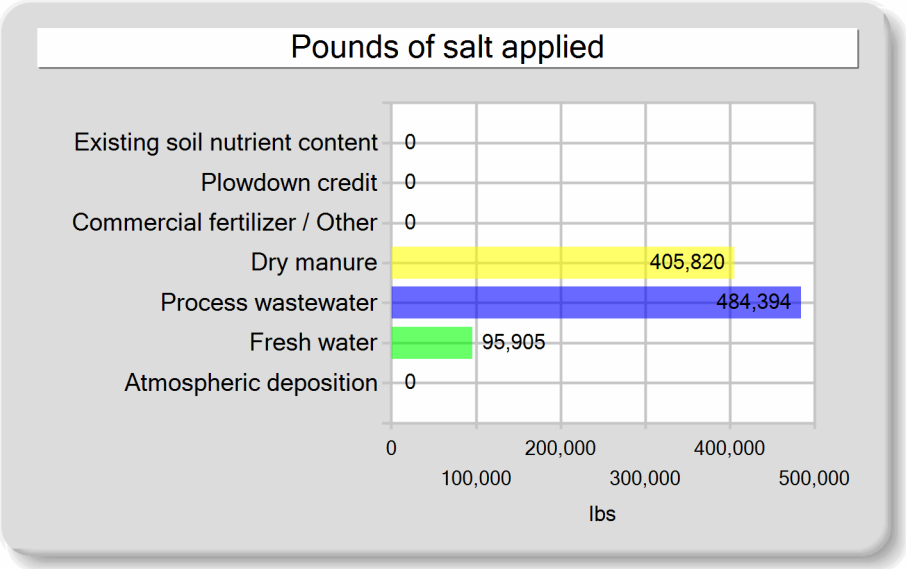


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Rice Field Oats (1.43) had high nitrogen field ratio as field failed to meet anticipated nitrogen removal of 180 pounds/acre. The subject field removed only 124.8 pounds of nitrogen per acre. Yield for the subject field was adequate however a lower than anticipated nitrogen value from the lab plant tissue analysis resulted in only 6.78 pounds of nitrogen removed per harvested ton compared to the anticipated 10 pounds of nitrogen removed per harvested ton resulting in the high nitrogen field ratio. Nitrogen application to the subject fields was not considered excessive but in line with that recommended in the dairy's nutrient management plan.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Rien Doornenbal
Rien Doornenbal (Jun 21, 2024 09:12 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Rien Doornenbal	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 21, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Rien Doornenbal

Name of Dairy Facility: Doornenbal Dairy #1

Facility Address: 19116 Lone Tree Road, Escalon, CA 95320
Number and Street City Zip Code

Contact Person Name and Phone Number: Rien Doornenbal 209-838-7108
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: Sousa's Custom Harvesting

Address of Hauling Company /Person: 20178 East Gawne Road, Stockton, CA 95215
Number and Street City Zip Code

Contact Person: Manuel Sousa 209-467-4122
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Sousa's Custom Harvesting, 20178 East Gawne Rd., Stockton, CA 95215 209-467-4122
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

20178 East Gawne Road, Stockton, CA 95215 185-100-03
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 11/13/2023

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 150 Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): 29.1% moisture
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

Hauler Calculations

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification E.3 of Reissued Waste Discharge Requirements General Order No. R5-2013-0122) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Rien Doornenbal Rien Doornenbal (Jun 21, 2024 09:12 PDT) Date: Jun 21, 2024

Hauler's Signature: _____ Date: _____

Sousa's Custom Harvesting
20178 East Gawne Road
Stockton, CA 95215

Date	Invoice #
11/28/2023	2482

Bill To

Doornenbal Dairy
P.O. Box 37
Escalon, CA 95320

Cattle Co.
Per Rien

Due Date
12/28/2023

Description	Quantity	Rate	Amount
Compost 2023			
15 loads (300 yards) of compost hauled 11/13/2023	15	125.00	1,875.00

Thank you for your business.

Total $1,875.00

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Rien Doornenbal

Name of Dairy Facility: Doornenbal Dairy #1

Facility Address: 19116 Lone Tree Road, Escalon, CA 95320
Number and Street City Zip Code

Contact Person Name and Phone Number: Rien Doornenbal 209-838-7108
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: Vander Meulen INC.

Address of Hauling Company /Person: 13282 Escalon Bellota Rd., Escalon, CA 95320
Number and Street City Zip Code

Contact Person: Steve Vander Meulen 209-838-3731
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Winters Farming Co. 18499 Jack Tone Road, Manteca, CA 95336 209-599-7502
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

18499 Jack Tone Road, Manteca, CA 95336 228-040-20
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 11/22/2023

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 98 Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): 29.1% moisture
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

Haulers Calculations

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification E.3 of Reissued Waste Discharge Requirements General Order No. R5-2013-0122) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Rien Doornenbal Rien Doornenbal (Jun 21, 2024 09:12 PDT) Date: Jun 21, 2024

Hauler's Signature: _____ Date: _____

VANDER MEULEN, INC.

13282 ESCALON BELLOTA RD
ESCALON, CA 95320

Invoice

Date	Invoice #
11/22/2023	1395

Bill To
R Doornenbal Dairy P.O. Box 37 Escalon, CA 95320

Customer

Date	Quantity	Item	Description
			COMPOST HAULING
	98	Hauling-Other	Doornenbal to Winters Farm Compost Hauling - 98 Tons Hauled

It's a pleasure working with you!

DELLAVALLE™
LABORATORY INC

R Doornenbal Dairy 1 & 3
19116 Lone Tree Rd
Escalon, CA 95320

Account# 00-0026366
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 12/14/2023 14:20
Reported: 12/21/2023 09:37

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0963-01	IW 3	Canal Water			12/13/2023 7:00
23L0963-02	IW 4	Canal Water			12/13/2023 7:00

Default Cooler Temperature on Receipt °C: 11.7
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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DELLAVALLE™
LABORATORY INC

R Doornenbal Dairy 1 & 3
19116 Lone Tree Rd
Escalon, CA 95320

Account# 00-0026366
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 12/14/2023 14:20
Reported: 12/21/2023 09:37

Sample Results

Sample: IW 3
23L0963-01 (Water)

Sampled: 12/13/2023 7:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.80**	mmhos/cm	0.01	1		12/18/23 16:17	SM 2510 B		BEL0700
Electrical Conductivity umhos	**801**	umhos/cm	10.0	1		12/18/23 16:17	SM 2510 B		BEL0700
Nitrate Nitrogen as NO3N	**19.5**	mg/L	0.1	1	10	12/15/23 11:45	EPA 300.0		BEL0595
Temperature	**25.0**	units	0.0	1		12/18/23 16:17	SM 4500-H+	H	BEL0700
pH	**7.7**	units	1.0	1		12/18/23 16:17	SM 4500-H+	H	BEL0700
Total Filterable Solids (TDS)	**490**	mg/L	10.0	1		12/19/23 14:37	SM 2540 C		BEL0669
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		12/19/23 14:30	SM 4500-NH3 C		BEL0671
Total Nitrogen	**19.8**	mg/L	1.00	1		12/19/23 14:30	SM 4500-NH3 C		BEL0671

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DELLAVALLE™
LABORATORY INC

R Doornenbal Dairy 1 & 3
19116 Lone Tree Rd
Escalon, CA 95320

Account# 00-0026366
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 12/14/2023 14:20
Reported: 12/21/2023 09:37

Sample Results
(Continued)

Sample: IW 4
23L0963-02 (Water)

Sampled: 12/13/2023 7:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.78**	mmhos/cm	0.01	1		12/18/23 16:18	SM 2510 B		BEL0700
Electrical Conductivity umhos	**776**	umhos/cm	10.0	1		12/18/23 16:18	SM 2510 B		BEL0700
Nitrate Nitrogen as NO3N	**23.9**	mg/L	0.1	1	10	12/15/23 12:05	EPA 300.0		BEL0595
Temperature	**25.0**	units	0.0	1		12/18/23 16:18	SM 4500-H+	H	BEL0700
pH	**7.8**	units	1.0	1		12/18/23 16:18	SM 4500-H+	H	BEL0700
Total Filterable Solids (TDS)	**510**	mg/L	10.0	1		12/19/23 14:37	SM 2540 C		BEL0669
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		12/19/23 14:47	SM 4500-NH3 C		BEL0673
Total Nitrogen	**24.2**	mg/L	1.00	1		12/19/23 14:47	SM 4500-NH3 C		BEL0673

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12/14/23 14:20 23L0963

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6120 800 228-9896 · Fax 559 268-8174

Purchase Order No | Bill To: ~~0~~ Acct # 26366 | Cons # 13

No. Samples: 19 | No of Bottles: 38

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater [ ] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: R Doornenbal Dairy 1&3

Other:

Address: 19116 Lone Tree Rd

City: Escalon State: CA Zip: 95320

Telephone: 209 838-7108 Fax:

Cell/Email:

COPY TO: F&A Ag Services
& JSawyer@Fragservices.com

REQUESTED BY: Jeanette Sawyer

PROJECT:

CROP:

Analysis and Bottles Required: (Please indicate Analysis)

DCW1 TTU

[ ] Co. Health Dept
[ ] RWQCB [ ] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Sampled By:

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field EC
1	IW 1 No sample Non/op				
2	↓ 2 ↓ ↓ ↓				
3	IW 3	12/13/23	7AM	1.7	
4	↓ 4	↓	↓	1.9	
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Jeanette Sawyer	F&R Ag Services		
Second	JLF	DLV	12-14-23 10:00	12-14-23 2:5pm
Third				
Fourth		DLV	12/14 14:2	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $3.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*caf*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *caf* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling hrs
Miles
Consulting

Shipping
$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature
Sample received in cooler with ice (coolant)
[ ] Yes [ ] No

pH Strips
Lot 10BDH4501
Exp. Jan. 2025

DELLAVALLE™
LABORATORY INC

R Doornenbal Dairy 1 & 3
19116 Lone Tree Rd
Escalon, CA 95320

Account# 00-0026366
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:02

Sample Results
(Continued)

Sample: DW Farea
23L0964-03 (Water)

Sampled: 12/13/2023 7:30
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.68**	mmhos/cm	0.01	1		12/18/23 16:29	SM 2510 B		BEL0700
Electrical Conductivity umhos	**683**	umhos/cm	10.0	1		12/18/23 16:29	SM 2510 B		BEL0700
Ammonia (as N)	*	mg/L	0.00	1		12/14/23 17:35	Field		BEL0647
Nitrate Nitrogen as NO3N	**14.6**	mg/L	0.1	1	10	12/16/23 19:48	EPA 300.0		BEL0665
Temperature	**25.0**	units	0.0	1		12/18/23 16:29	SM 4500-H+	H	BEL0700
pH	**7.7**	units	1.0	1		12/18/23 16:29	SM 4500-H+	H	BEL0700

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DELLAVALLE™
LABORATORY INC

R Doornenbal Dairy 1 & 3
19116 Lone Tree Rd
Escalon, CA 95320

Account# 00-0026366
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:02

Sample Results
(Continued)

Sample: DW Farea Moble 8275
23L0964-04 (Water)

Sampled: 12/13/2023 7:30
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.69**	mmhos/cm	0.01	1		12/18/23 16:31	SM 2510 B		BEL0700
Electrical Conductivity umhos	**686**	umhos/cm	10.0	1		12/18/23 16:31	SM 2510 B		BEL0700
Ammonia (as N)	*	mg/L	0.00	1		12/14/23 17:35	Field		BEL0647
Nitrate Nitrogen as NO3N	**14.7**	mg/L	0.1	1	10	12/16/23 20:08	EPA 300.0		BEL0665
Temperature	**25.0**	units	0.0	1		12/18/23 16:31	SM 4500-H+	H	BEL0700
pH	**7.9**	units	1.0	1		12/18/23 16:31	SM 4500-H+	H	BEL0700

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DELLAVALLE™
LABORATORY INC

R Doornenbal Dairy 1 & 3
19116 Lone Tree Rd
Escalon, CA 95320

Account# 00-0026366
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:02

Sample Results
(Continued)

Sample: DW Main Dairy
23L0964-05 (Water)

Sampled: 12/13/2023 7:30
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.72**	mmhos/cm	0.01	1		12/18/23 16:32	SM 2510 B		BEL0700
Electrical Conductivity umhos	**717**	umhos/cm	10.0	1		12/18/23 16:32	SM 2510 B		BEL0700
Ammonia (as N)	*	mg/L	0.00	1		12/14/23 17:35	Field		BEL0647
Nitrate Nitrogen as NO3N	**16.3**	mg/L	0.1	1	10	12/16/23 20:29	EPA 300.0		BEL0665
Temperature	**25.0**	units	0.0	1		12/18/23 16:32	SM 4500-H+	H	BEL0700
pH	**7.9**	units	1.0	1		12/18/23 16:32	SM 4500-H+	H	BEL0700

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DELLAVALLE™
LABORATORY INC

R Doornenbal Dairy 1 & 3
19116 Lone Tree Rd
Escalon, CA 95320

Account# 00-0026366
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:02

Sample Results
(Continued)

Sample: DW Reins House
23L0964-06 (Water)

Sampled: 12/13/2023 7:30
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.29**	mmhos/cm	0.01	1		12/18/23 16:33	SM 2510 B		BEL0700
Electrical Conductivity umhos	**287**	umhos/cm	10.0	1		12/18/23 16:33	SM 2510 B		BEL0700
Ammonia (as N)	*	mg/L	0.00	1		12/14/23 17:35	Field		BEL0647
Nitrate Nitrogen as NO3N	**5.8**	mg/L	0.1	1	10	12/16/23 20:49	EPA 300.0		BEL0665
Temperature	**25.0**	units	0.0	1		12/18/23 16:33	SM 4500-H+	H	BEL0700
pH	**7.9**	units	1.0	1		12/18/23 16:33	SM 4500-H+	H	BEL0700

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DELLAVALLE™
LABORATORY INC

R Doornenbal Dairy 1 & 3
19116 Lone Tree Rd
Escalon, CA 95320

Account# 00-0026366
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:02

Sample Results
(Continued)

Sample: DW Roos
23L0964-07 (Water)

Sampled: 12/13/2023 7:30
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.57**	mmhos/cm	0.01	1		12/18/23 16:35	SM 2510 B		BEL0700
Electrical Conductivity umhos	**566**	umhos/cm	10.0	1		12/18/23 16:35	SM 2510 B		BEL0700
Ammonia (as N)	*	mg/L	0.00	1		12/14/23 17:35	Field		BEL0647
Nitrate Nitrogen as NO3N	**10.4**	mg/L	0.1	1	10	12/16/23 21:09	EPA 300.0		BEL0665
Temperature	**25.0**	units	0.0	1		12/18/23 16:35	SM 4500-H+	H	BEL0700
pH	**7.8**	units	1.0	1		12/18/23 16:35	SM 4500-H+	H	BEL0700

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DELLAVALLE™
LABORATORY INC

R Doornenbal Dairy 1 & 3
19116 Lone Tree Rd
Escalon, CA 95320

Account# 00-0026366
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:02

Sample Results
(Continued)

Sample: DW Tony's
23L0964-08 (Water)

Sampled: 12/13/2023 7:30
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.54**	mmhos/cm	0.01	1		12/18/23 16:36	SM 2510 B		BEL0700
Electrical Conductivity umhos	**538**	umhos/cm	10.0	1		12/18/23 16:36	SM 2510 B		BEL0700
Ammonia (as N)	*	mg/L	0.00	1		12/14/23 17:35	Field		BEL0647
Nitrate Nitrogen as NO3N	**9.6**	mg/L	0.1	1	10	12/16/23 21:29	EPA 300.0		BEL0665
Temperature	**25.0**	units	0.0	1		12/18/23 16:36	SM 4500-H+	H	BEL0700
pH	**7.8**	units	1.0	1		12/18/23 16:36	SM 4500-H+	H	BEL0700

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

R Doornenbal Dairy 1 & 3
19116 Lone Tree Rd
Escalon, CA 95320

Account# 00-0026366
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:02

Sample Results
(Continued)

Sample: DW Zumwalt
23L0964-09 (Water)

Sampled: 12/13/2023 7:30
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.71**	mmhos/cm	0.01	1		12/18/23 16:37	SM 2510 B		BEL0700
Electrical Conductivity umhos	**706**	umhos/cm	10.0	1		12/18/23 16:37	SM 2510 B		BEL0700
Ammonia (as N)	*	mg/L	0.00	1		12/14/23 17:35	Field		BEL0647
Nitrate Nitrogen as NO3N	**25.0**	mg/L	0.1	1	10	12/16/23 21:50	EPA 300.0		BEL0665
Temperature	**25.0**	units	0.0	1		12/18/23 16:37	SM 4500-H+	H	BEL0700
pH	**7.8**	units	1.0	1		12/18/23 16:37	SM 4500-H+	H	BEL0700

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12/14/23 14:20 23L0964

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

Purchase Order No | Bill To: Acct # ~~0~~ 26366 Cons # 13 | No. Samples: 9 | No of Bottles: 9

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater [ ] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: R Doornenbal Dairy 1&3
Address: 19116 Lone Tree Rd
City: Escalon State: CA Zip: 95320
Telephone: 209 838-7108 Fax:
Cell/Email:
COPY TO: F&A Ag Services
& JSawyer@Fragservices.com
REQUESTED BY: Jeanette Sawyer
PROJECT:
CROP:

Other:

Analysis and Bottles Required: (*Please indicate Analysis*)
DWW1

[ ] Co. Health Dept
[ ] RWQCB [ ] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Sampled By:

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field EC
1	DW Dairy 3 New	12/13/23	Started 7:30	11.7	
2	Dairy 3 old			12.7	
3	Faria			10.0	
4	Faria Mobile 8275			12.4	
5	Main Dairy			10.8	
6	Reins House			9.8	
7	Ross			9.8	
8	Tony's			10.0	
9	Zumwalt			11.6	
10					

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Jeanette Sawyer	F&R Ag Services		
Second	JLF	DLV	12-14-23 10:00	12-14-23 2:15 pm
Third				
Fourth			12/14 14:20	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:
Sampling hrs
Miles
Consulting

Shipping
$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature
Sample received in cooler with ice (coolant)
[ ] Yes [ ] No

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

2023 Annual Report Doornenbal Dairy #1

Final Audit Report 2024-06-21

Created:	2024-06-20
By:	Vince Furtado (vfurtado@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAvZjei6kGrSgV_sPN33FU-H56vVED8Q9U

"2023 Annual Report Doornenbal Dairy #1" History

Document created by Vince Furtado (vfurtado@fragservices.com)
2024-06-20 - 4:28:44 PM GMT

Document emailed to mbrasil@rdranches.com for signature
2024-06-20 - 4:30:56 PM GMT

Email viewed by mbrasil@rdranches.com
2024-06-21 - 4:12:20 PM GMT

Signer mbrasil@rdranches.com entered name at signing as Rien Doornenbal
2024-06-21 - 4:12:53 PM GMT

Document e-signed by Rien Doornenbal (mbrasil@rdranches.com)
Signature Date: 2024-06-21 - 4:12:55 PM GMT - Time Source: server

Agreement completed.
2024-06-21 - 4:12:55 PM GMT